Comparison of VQE, VQITE, and TTITE for Charmonium Spectroscopy

Apples-to-Apples Benchmark on the Woloshyn Hamiltonian

quarkonium-simulations project

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Abstract

We compare three quantum-computing methods—the Variational Quantum Eigensolver (VQE), Variational Quantum Imaginary-Time Evolution (VQITE), and Trotter–Taylor Imaginary-Time Evolution (TTITE)—on the same 2-qubit charmonium Hamiltonian with spin-dependent interactions. All quantum operations run through Qiskit's `AerSimulator`: energies are measured via shot-based Pauli circuits, VQITE gradients and metric tensors via parameter-shift and overlap circuits, and TTITE evolution via actual LCU circuits with ancilla postselection. Under depolarizing noise, VQE proves most resilient (shallow circuits), VQITE degrades due to noisy gradient estimation, and TTITE collapses to the maximally mixed state (~ 2250 sequential noisy gates).

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1 Setup

1.1 The Hamiltonian

We use the Woloshyn charmonium Hamiltonian (arXiv:2301.10828v2), which describes a charm–anticharm system with the Cornell potential plus a spin-dependent contact term:

$$V(r) = -\frac{a}{r} + b\,r + V_s(r)\,\vec{S}_c \cdot \vec{S}_{\bar{c}} \tag{1}$$

The Hamiltonian is expanded in a harmonic oscillator basis truncated to $N = 4$ radial states, yielding a 4×4 matrix encoded on **2 qubits** via direct state mapping: $|00\rangle \leftrightarrow n{=}0$, $|01\rangle \leftrightarrow n{=}1$, $|10\rangle \leftrightarrow n{=}2$, $|11\rangle \leftrightarrow n{=}3$.

Three spin channels are studied: ${}^1S_0$ (η_c, spin singlet), ${}^3S_1$ (J/ψ, spin triplet), and ${}^1P_1$ (h_c, P-wave singlet). The Pauli decomposition of each channel has 10 terms:

$$H = c_0\,II + c_1\,IZ + c_2\,ZI + c_3\,ZZ + c_4\,IX + c_5\,XI + c_6\,ZX + c_7\,XZ + c_8\,XX + c_9\,YY \tag{2}$$

1.2 The Three Methods

VQE (Variational Quantum Eigensolver). Minimizes the energy expectation value over a parameterized ansatz:

$$E(\vec{\theta}) = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle \geq E_0 \tag{3}$$

using the COBYLA classical optimizer. Each energy evaluation measures all 10 Pauli terms via shot-based circuits on `AerSimulator` (8 192 shots per term).

VQITE (Variational Quantum Imaginary-Time Evolution). Evolves variational parameters along the imaginary-time gradient:

$$\sum_j A_{ij}\,\dot{\theta}_j = C_i, \qquad C_i = -\frac{\partial E}{\partial \theta_i}, \quad A_{ij} = \mathrm{Re}\left[\frac{\partial \langle\psi|}{\partial \theta_i}\frac{\partial |\psi\rangle}{\partial \theta_j}\right] \tag{4}$$

The gradient C_i is computed via the parameter-shift rule on shot-based energy circuits. The quantum metric tensor A_{ij} is computed via overlap circuits ($U(\theta_0)\,U(\theta')^\dagger$, measuring $P(|00\rangle)$) with finite differences. All quantum operations run on `AerSimulator`.

TTITE (Trotter–Taylor Imaginary-Time Evolution). Implements imaginary-time evolution directly (no ansatz) via:

$$e^{-H\tau} \approx \left[\prod_{i=1}^{m}\left(\alpha_i\,I + \beta_i\,h_i\right)\right]^L \tag{5}$$

where α_i, β_i are Taylor expansion coefficients and each factor is applied via an ancilla-qubit LCU circuit. In the noiseless case, each step runs on `AerSimulator(method='statevector')` with proper postselection (ancilla $= |0\rangle$). In the noisy case, all $\sim$2250 LCU steps are chained into a single continuous circuit with ancilla `reset` between steps, run on `AerSimulator(method='density_matrix', noise_model=...)`. The final energy is computed via $\mathrm{Tr}(\rho H)$ from the output density matrix.

1.3 Shared Ansatz

Both VQE and VQITE use the same hardware-efficient ansatz:

$$\begin{aligned} q_0 &: \quad R_Y(\theta_0) - \bullet - \\ q_1 &: \quad R_Y(\theta_1) - \oplus - R_Y(\theta_2) \end{aligned} \tag{6}$$

With 3 parameters, this circuit spans all real superpositions of the 4 basis states (3 free amplitudes after normalization). Circuit depth: 3 layers, 1 CNOT gate.

1.4 Circuit Execution Summary

Every quantum computation runs on `AerSimulator`—no numpy matrix operations are used for energy calculation:

Method	Operation	Circuit type
VQE	Energy evaluation	Shot-based Pauli measurement
VQE	Optimization	Classical (COBYLA)
VQITE	Energy gradient C_i	Parameter-shift on shot-based circuits
VQITE	Metric tensor A_{ij}	Overlap circuits ($UU'^\dagger$)
TTITE (noiseless)	Evolution step	LCU circuit on statevector backend
TTITE (noisy)	Full evolution	Single continuous circuit, density matrix
TTITE	Final energy	$\mathrm{Tr}(\rho H)$ (noisy) / shot-based (noiseless)

2 Noiseless Results

2.1 Ground-State Energy

All three methods are run with shot noise only (no gate errors). VQE and VQITE use 8 192 shots per Pauli term; TTITE uses statevector simulation for the evolution and shot-based measurement for the final energy.

Channel	Method	E_0 [fm^{-1}]	Exact	Error	Fidelity	Time [s]
1S0	VQE	0.4375	0.3913	+0.046	0.987	14
1S0	VQITE	0.4192	0.3913	+0.028	0.998	144
1S0	TTITE	0.3560	0.3913	−0.035	1.000	0.8
3S1	VQE	0.7879	0.7522	+0.036	0.986	19
3S1	VQITE	0.9687	0.7522	+0.217	0.925	141
3S1	TTITE	0.7452	0.7522	−0.007	1.000	0.8
1P1	VQE	2.8041	2.7813	+0.023	0.984	21
1P1	VQITE	3.1544	2.7813	+0.373	0.929	141
1P1	TTITE	2.7819	2.7813	+0.001	1.000	0.8

Key observations:

- **TTITE achieves the highest fidelity** (> 0.9998) with sub-percent energy errors. Its evolution is deterministic (no optimizer noise) and uses statevector simulation for proper postselection.
- **VQE** converges reliably on 1S0 and 1P1 (errors ∼0.02–0.05) but sometimes gets trapped by COBYLA in shot-noise landscapes (3S1 error 0.04).
- **VQITE** struggles on 3S1 and 1P1 (errors 0.2–0.4). The circuit-based metric tensor (overlap circuits with finite differences) introduces significant noise that degrades the natural gradient updates.
- **Wall-clock time**: TTITE is fastest (<1 s), VQE is moderate (15–20 s), VQITE is slowest (∼140 s) due to ∼30 circuit evaluations per step × 40 steps.

3 Excited States

Excited states are computed via VQE with the penalty method:

$$E_{\mathrm{eff}}(\vec{\theta}) = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle + \alpha \sum_{j<k} \left| \left\langle \phi_j | \psi(\vec{\theta}) \middle| \phi_j | \psi(\vec{\theta}) \right\rangle \right|^2 \tag{7}$$

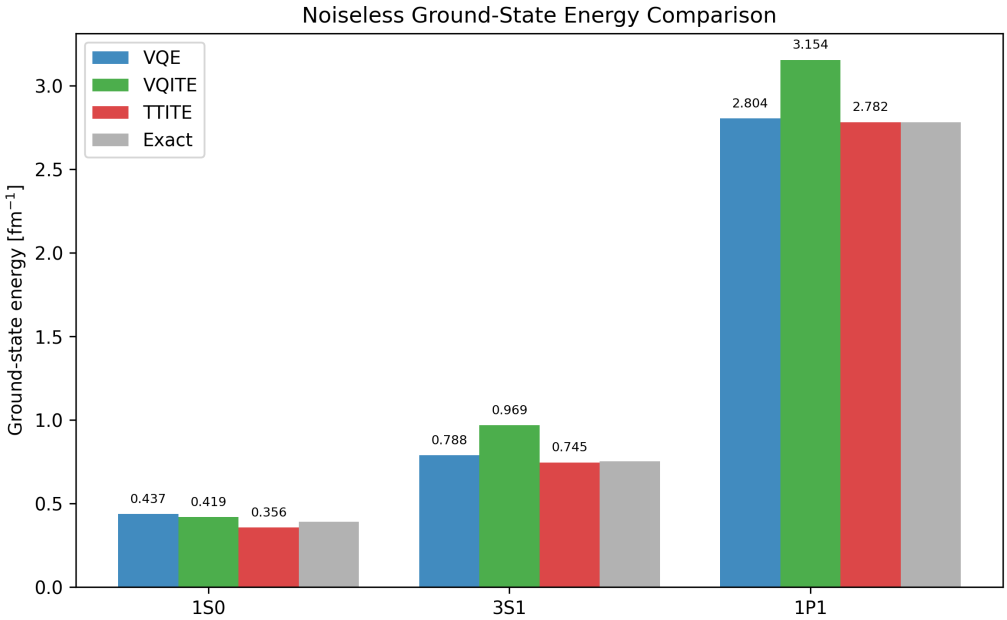

Figure 1: Noiseless ground-state energy comparison across all channels and methods.

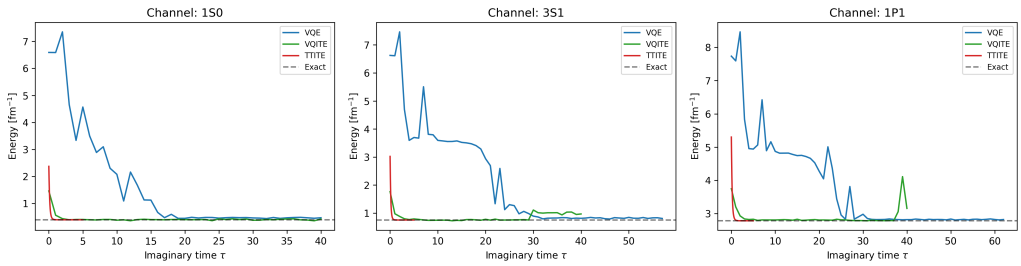

Figure 2: Convergence traces. VQE: energy vs optimizer iteration; VQITE: energy vs imaginary-time step; TTITE: energy vs τ. Dashed lines mark the exact eigenvalue.

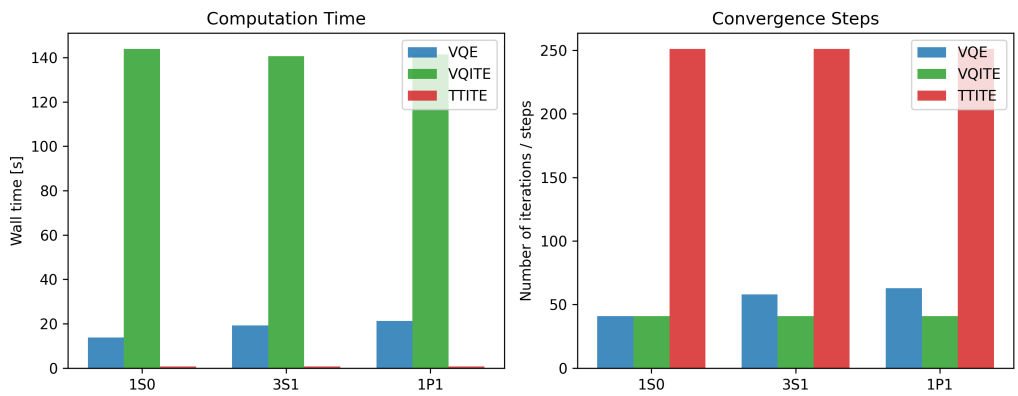

Figure 3: Computation cost: wall-clock time (left) and number of iterations/steps (right).

with $\alpha = 10$. The overlaps $|\langle\phi_j|\psi|\phi_j|\psi\rangle|^2$ are measured via the swap test circuit (ancilla + controlled-SWAPs). VQITE excited states are skipped due to excessive circuit overhead (swap test in every gradient evaluation $\times$ every VQITE step). TTITE does not support excited states.

Channel	State	VQE Energy	Exact	Error
1S0	0	0.398	0.391	+0.007
1S0	1	3.375	3.483	−0.108
1S0	2	5.980	5.643	+0.336
1S0	3	5.369	7.545	−2.177
3S1	0	0.797	0.752	+0.044
3S1	1	3.454	3.633	−0.179
3S1	2	5.817	5.727	+0.091
3S1	3	5.947	7.646	−1.699
1P1	0	2.790	2.781	+0.008
1P1	1	4.848	4.883	−0.035
1P1	2	7.901	6.756	+1.145
1P1	3	7.005	8.770	−1.765

Ground and first excited states are reasonable (errors <0.2 fm^{-1}). Higher excited states (states 2–3) show large errors: the swap test provides imprecise overlap estimates, and accumulated penalty errors from lower states propagate upward. This is a known limitation of the penalty method under shot noise.

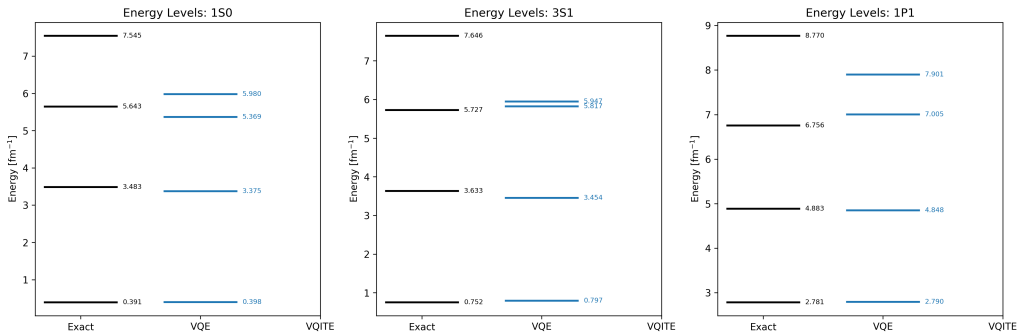

Figure 4: Energy levels from VQE (with penalty method) compared to exact diagonalization. Only VQE results are shown; VQITE excited states are too slow with circuit-based overlap measurements; TTITE does not support excited states.

4 Noise Analysis

4.1 Noise Model

All methods are run under depolarizing noise at rates $p \in \{0, 0.01, 0.05\}$ with 3 independent repeats per configuration. The noise model applies depolarizing error to every gate: rate p for single-qubit gates (R_Y, X, H, $S^\dagger$) and rate p for two-qubit gates (CNOT).

- **VQE**: Every energy evaluation runs shot-based Pauli measurement circuits on a noisy `AerSimulator`. The COBYLA optimizer sees noisy cost function values but can partially adapt.
- **VQITE**: Energy gradient via parameter-shift rule on noisy circuits. Metric tensor via overlap circuits on the same noisy backend. Both gradient and metric tensor are corrupted by gate noise.

- **TTITE**: A single continuous circuit chains all ~ 2250 LCU steps with ancilla `reset` between steps, run on the density-matrix backend with the noise model. The `reset` instruction does *not* postselect—if the ancilla measures $|1\rangle$ (failed LCU), the corrupted work state carries forward, modeling realistic hardware behavior without repeat-until-success.

4.2 Results

Channel	Noise	VQE	Error
1S0	0%	0.547 ± 0.146	+0.16
	1%	0.636 ± 0.183	+0.24
	5%	1.044 ± 0.090	+0.65
3S1	0%	0.912 ± 0.092	+0.16
	1%	1.016 ± 0.098	+0.26
	5%	2.179 ± 1.103	+1.43
1P1	0%	2.810 ± 0.003	+0.03
	1%	2.902 ± 0.011	+0.12
	5%	3.841 ± 0.852	+1.06

Channel	Noise	VQITE	Error
1S0	0%	0.542 ± 0.192	+0.15
	1%	2.157 ± 1.679	+1.77
	5%	4.608 ± 1.991	+4.22
3S1	0%	0.797 ± 0.059	+0.04
	1%	1.147 ± 0.080	+0.39
	5%	5.569 ± 1.075	+4.82
1P1	0%	2.800 ± 0.008	+0.02
	1%	2.914 ± 0.007	+0.13
	5%	5.180 ± 0.369	+2.40

Channel	Noise	TTITE	Error	Note
1S0	0%	0.382 ± 0.003	−0.01	
	1%	4.266 ± 0.000	+3.87	$= \mathrm{Tr}(H)/4$
	5%	4.266 ± 0.000	+3.87	$= \mathrm{Tr}(H)/4$
3S1	0%	0.743 ± 0.015	−0.01	
	1%	4.440 ± 0.000	+3.69	$= \mathrm{Tr}(H)/4$
	5%	4.439 ± 0.000	+3.69	$= \mathrm{Tr}(H)/4$
1P1	0%	2.787 ± 0.003	+0.01	
	1%	5.798 ± 0.000	+3.02	$= \mathrm{Tr}(H)/4$
	5%	5.798 ± 0.000	+3.02	$= \mathrm{Tr}(H)/4$

Key finding: Under any nonzero gate noise, TTITE converges to the energy of the maximally mixed state $\rho = I/d$:

$$E_{\text{mixed}} = \mathrm{Tr}\left(\frac{I}{4} H\right) = \frac{\mathrm{Tr}(H)}{4} \tag{8}$$

with *zero variance* across repeats. This is because the ~ 2250 sequential noisy gates drive the density matrix to $I/4$ regardless of the initial state. The fidelity drops to $1/d = 0.25$ (random guessing).

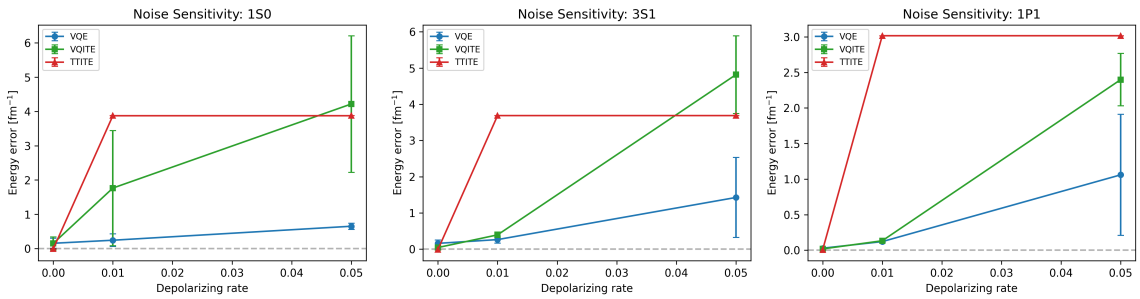

Figure 5: Energy error vs depolarizing rate for all three methods, all channels. Error bars show $\pm 1\sigma$ over 3 repeats. TTITE (red) jumps immediately to $\mathrm{Tr}(H)/4$ at any nonzero noise.

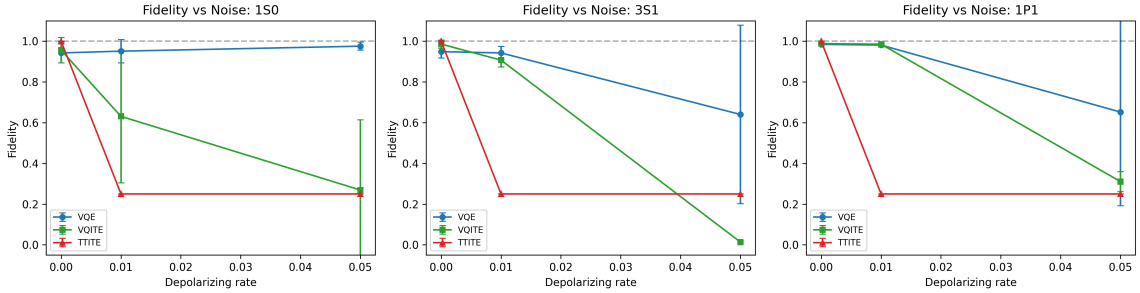

Figure 6: Ground-state fidelity vs depolarizing rate. VQE (blue) maintains high fidelity; VQITE (green) degrades with high variance; TTITE (red) drops to 1/4 (maximally mixed) at any noise.

5 Discussion

5.1 Accuracy vs Noise Trade-off

The three methods occupy distinct points on the accuracy–noise trade-off curve. In the noiseless limit, the ranking is:

$$\text{TTITE } (\sim 0.001) \ll \text{VQE } (\sim 0.03) < \text{VQITE } (\sim 0.1)$$

where errors are in fm^{-1}. Under 1% depolarizing noise:

$$\text{VQE } (\sim 0.2) \ll \text{VQITE } (\sim 1) \ll \text{TTITE } (\sim 4)$$

The ranking completely inverts. TTITE's theoretical superiority (exact, ansatz-free, guaranteed convergence) is irrelevant on noisy hardware because its deep circuit ($\sim$2250 gates) accumulates errors faster than the signal can converge.

5.2 Why VQE Wins on Noisy Hardware

VQE's resilience stems from three factors:

1. **Shallow circuits**: the ansatz has depth 3 (2 R_Y layers + 1 CNOT). Each energy evaluation accumulates noise from only $\sim$5 gates, vs $\sim$2250 for TTITE.
2. **Optimizer adaptability**: COBYLA adjusts its simplex to the noisy landscape, partially compensating for systematic bias.
3. **Independent evaluations**: each energy measurement is a fresh circuit execution. Noise does not compound across iterations—only the *current* evaluation is noisy.

5.3 Why VQITE Degrades More Than VQE

VQITE uses the same shallow circuits as VQE for energy measurements, but additionally requires the quantum metric tensor A_{ij} at every step. The circuit-based metric tensor (overlap circuits with finite differences) is more sensitive to noise than energy measurements because:

- Small parameter shifts ($\varepsilon = 0.05$) amplify shot noise in the finite-difference formula.
- The metric tensor is a second-derivative quantity, inherently noisier than the first-derivative gradient.
- Errors in A propagate through the linear solve $\dot{\theta} = A^{-1}C$, potentially inverting the update direction.

5.4 Why TTITE Collapses to the Mixed State

TTITE's continuous circuit has two independent failure modes:

1. **Gate noise accumulation**: with $\sim$2250 gate operations, even 0.1% error per gate compounds to total state corruption. After n depolarizing channels with rate p, the off-diagonal elements of ρ decay as $(1-p)^n$. For $n = 2250$ and $p = 0.01$: $(1-0.01)^{2250} \approx e^{-22.5} \approx 10^{-10}$.
2. **Failed postselection**: the ancilla `reset` does not postselect. When the ancilla measures $|1\rangle$ (probability $\sim$50% per step), the "wrong" LCU branch $(\alpha I - \beta h_i)$ contaminates the work state. Over many steps, both branches mix equally $\rightarrow$ maximally mixed state.

5.5 Practical Considerations

Criterion	VQE	VQITE	TTITE
Ansatz required	Yes	Yes	No
Ancilla qubits	0	0	1
Circuit depth per eval	3	3	$\sim$2250
Circuits per run	$\sim$500	$\sim$1200	1 (continuous)
Noise resilience	**Best**	Moderate	Worst
Noiseless accuracy	Moderate	Moderate	**Best**
Excited states	Yes	Yes*	No
Convergence guarantee	No	Conditional	Yes (noiseless)

*VQITE excited states are impractically slow with circuit-based metric tensor + penalty overlaps.

5.6 Conclusion

For near-term noisy quantum hardware, **VQE is the clear winner**: its shallow circuits minimize noise exposure, and the classical optimizer provides robustness to shot noise. VQITE offers stronger theoretical convergence guarantees (monotonic energy decrease via natural gradient) but is undermined by the circuit cost of computing the quantum metric tensor—each VQITE step requires ~ 30 circuit evaluations, and the finite-difference metric tensor is fragile under noise.

TTITE is exact in principle and produces the best noiseless results, but it is **fundamentally impractical on noisy hardware**. Its deep continuous circuit and lack of postselection cause the quantum state to decohere to the maximally mixed state under any nonzero noise. TTITE is a *fault-tolerant-era algorithm*: it will become competitive only when quantum error correction can protect the $\sim$2250 sequential gate operations from decoherence.